

Executive Summary: Bhopal Delivery Turnaround Time (TAT) & SLA Analysis

1. Executive Summary & Operational Context

An end-to-end analysis of **5,000 delivery orders across 20 localities in Bhopal** was conducted to evaluate delivery turnaround times (TAT), platform performance (Zomato vs. Swiggy), and Service Level Agreement (SLA) breach rates.

The overall average delivery TAT stands at **39.6 minutes**. Granular SQL analysis reveals significant localized SLA failures, with breach rates exceeding **80%** in several areas, particularly during high-demand periods.

2. Key SQL Findings & Operational Bottlenecks

A. Localized SLA Breach Crisis

- **Highest SLA Breach Areas:**
Indrapuri (84.83%), **Shahpura (83.77%)**, and **Berasia Road (83.33%)** recorded the highest proportion of orders exceeding the 30-minute delivery threshold.
- **Highest Average TAT:**
Bairagarh (40.80 mins) and **Berasia Road (40.75 mins)** recorded the highest average delivery times among the 20 localities.

B. Hourly Delays & Peak Demand Windows

- Delivery delays peak during **late-morning and mid-day dispatch windows**, with **10:00 AM recording an average TAT of 41.77 minutes**, alongside elevated TAT during the evening rush between **7:00 PM and 10:00 PM**.
- **Average Order Value (AOV)** remains relatively stable between **₹275 and ₹298** during peak delay hours, suggesting that differences in basket value alone do not explain the observed delivery delays.

C. Platform Delivery Performance

- **Swiggy:**
Bairagarh (41.74 mins) and **Shahpura (40.96 mins)** recorded the highest average TAT among the analyzed Swiggy locality segments.
- **Zomato:**
Berasia Road (41.31 mins) and **Govindpura (40.93 mins)** recorded the highest average TAT among the analyzed Zomato locality segments.

3. Strategic Action Plan for Quick-Commerce Operations

1. Dark Store / Hub Expansion in High-Breach Zones

Problem:

Indrapuri, Shahpura, and Berasia Road show SLA breach rates above **82%**.

Action:

Evaluate the feasibility of establishing micro-fulfillment dark stores or dedicated rider hubs within approximately **2 km** of high-breach zones to potentially reduce delivery distances and transit times.

2. Dynamic Rider Re-allocation During Peak Windows

Action:

Reallocate available off-peak delivery capacity toward **Bairagarh and Berasia Road** during the identified high-delay windows:

- **10:00 AM – 12:00 PM**
- **7:00 PM – 10:00 PM**

A pilot allocation of approximately **20% of available idle capacity** could be evaluated against baseline TAT and SLA performance.

3. Merchant Kitchen SLA & Batch Dispatching

Action:

Implement automated preparation-time alerts for merchants or cuisine categories associated with higher TAT, particularly **North Indian and Chinese**, and coordinate rider dispatching with expected food readiness times.

Key Takeaway

The analysis indicates that delivery performance varies substantially by **locality, time of day, and platform**. The largest operational opportunities identified are concentrated around **high-SLA-breach localities and peak-hour capacity management**.